

KPI Documentation

Apex Manufacturing Executive Dashboard

Project Name: Apex Manufacturing Executive Dashboard
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1. Purpose

This document defines all Key Performance Indicators (KPIs) used in the Apex Manufacturing Executive Dashboard. It provides a standardized reference for business users, analysts, and developers by explaining each KPI's definition, business value, calculation logic, formatting, and reporting purpose.

2. KPI Summary

KPI	Category	Format
Total Production	Production	Whole Number
Production Target	Production	Whole Number
Production Efficiency	Production	Percentage
Defective Units	Quality	Whole Number
Defect Rate	Quality	Percentage
Inventory Stock	Inventory	Whole Number
Maintenance Cost	Maintenance	Currency
Purchase Orders	Procurement	Whole Number
Procurement Cost	Procurement	Currency
Active Suppliers	Procurement	Whole Number

3. KPI Definitions

3.1 Total Production

Attribute	Details
KPI Name	Total Production
Business Purpose	Measures the total number of units manufactured during the selected period.
Category	Production
Business Formula	Sum of all produced units
Example DAX Measure	<code>SUM(Production[Quantity_Produced])</code>
Format	Whole Number
Higher Value	Better

3.2 Production Target

Attribute	Details
KPI Name	Production Target
Business Purpose	Represents the planned production quantity for the selected period.
Category	Production
Business Formula	Sum of target quantities
Example DAX Measure	<code>SUM(Production[Target_Quantity])</code>
Format	Whole Number
Higher Value	Better

3.3 Production Efficiency

Attribute	Details
KPI Name	Production Efficiency
Business Purpose	Measures production performance against the planned target.
Category	Production
Business Formula	$(\text{Total Production} \div \text{Production Target}) \times 100$

Attribute	Details
Example DAX Measure	<code>DIVIDE([Total Production],[Production Target],0)</code>
Format	Percentage
Target	100% or Above

3.4 Defective Units

Attribute	Details
KPI Name	Defective Units
Business Purpose	Total number of products identified as defective during inspection.
Category	Quality
Business Formula	Sum of defective units
Example DAX Measure	<code>SUM(Quality[Defective_Units])</code>
Format	Whole Number
Lower Value	Better

3.5 Defect Rate

Attribute	Details
KPI Name	Defect Rate
Business Purpose	Indicates the percentage of production that failed quality inspection.
Category	Quality
Business Formula	$(\text{Defective Units} \div \text{Total Production}) \times 100$
Example DAX Measure	<code>DIVIDE([Defective Units],[Total Production],0)</code>
Format	Percentage
Lower Value	Better

3.6 Inventory Stock

Attribute	Details
KPI Name	Inventory Stock
Business Purpose	Displays the total available inventory across all warehouses.

Attribute	Details
Category	Inventory
Business Formula	Sum of available stock
Example DAX Measure	<code>SUM(Inventory[Available_Stock])</code>
Format	Whole Number
Business Goal	Maintain optimal stock levels

3.7 Maintenance Cost

Attribute	Details
KPI Name	Maintenance Cost
Business Purpose	Measures total expenditure on equipment maintenance.
Category	Maintenance
Business Formula	Sum of maintenance costs
Example DAX Measure	<code>SUM(Maintenance[Maintenance_Cost])</code>
Format	Currency
Lower Value	Better (without impacting reliability)

3.8 Purchase Orders

Attribute	Details
KPI Name	Purchase Orders
Business Purpose	Counts procurement transactions completed during the selected period.
Category	Procurement
Business Formula	Count of purchase orders
Example DAX Measure	<code>COUNT(Purchase_Orders[PO_ID])</code>
Format	Whole Number

3.9 Procurement Cost

Attribute	Details
KPI Name	Procurement Cost

Attribute	Details
Business Purpose	Measures total purchasing expenditure.
Category	Procurement
Business Formula	Sum of procurement costs
Example DAX Measure	<code>SUM(Purchase_Orders[Total_Cost])</code>
Format	Currency
Business Goal	Optimize purchasing spend

3.10 Active Suppliers

Attribute	Details
KPI Name	Active Suppliers
Business Purpose	Counts the number of suppliers contributing to procurement activities.
Category	Procurement
Business Formula	Distinct count of suppliers
Example DAX Measure	<code>DISTINCTCOUNT(Suppliers[Supplier_ID])</code>
Format	Whole Number

4. KPI Interaction

All KPI cards are dynamic and respond to dashboard filters and slicers, including:

- Date
- Plant
- Product
- Machine
- Department (if applicable)

This allows users to analyze operational performance for specific business segments.

5. KPI Design Standards

The following standards are applied across the dashboard:

- Whole numbers displayed with thousand separators.
- Currency formatted consistently.
- Percentages displayed with two decimal places.

- Dynamic updates based on user selections.
 - Business-friendly KPI names.
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6. Business Value

The KPI framework enables stakeholders to:

- Monitor production performance.
 - Evaluate manufacturing efficiency.
 - Detect quality issues.
 - Track maintenance expenditure.
 - Monitor inventory levels.
 - Analyze procurement activity.
 - Support operational and strategic decision-making.
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7. Future KPI Enhancements

Potential KPIs for future versions include:

- Overall Equipment Effectiveness (OEE)
 - Machine Utilization Rate
 - Downtime Percentage
 - Inventory Turnover Ratio
 - Supplier On-Time Delivery Rate
 - Mean Time Between Failures (MTBF)
 - Mean Time to Repair (MTTR)
 - Production Cost per Unit
 - Procurement Lead Time
 - Forecast vs Actual Production
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8. Conclusion

The KPIs documented in this report provide a consistent framework for monitoring manufacturing performance across production, quality, inventory, maintenance, and procurement functions. By standardizing definitions and calculation logic, the dashboard ensures reliable reporting, supports informed decision-making, and promotes alignment between business stakeholders and technical teams.